

6	(a)	(i)	$C_0 = 65 \text{ s}^{-1}$ $C = C_0 e^{-\lambda t} = C_0 e^{-(\ln 2 / t_{1/2}) t}$ When $t = 46 \text{ s}$, $C = 35 \text{ s}^{-1} \Rightarrow 35 = 65 e^{-(\ln 2 / t_{1/2})(46)} \Rightarrow t_{1/2} = 51.5 \text{ s}$ B1 for reading values correctly from graph, A1 for 52 s When $t = 110 \text{ s}$, $C = 15 \text{ s}^{-1} \Rightarrow 15 = 65 e^{-(\ln 2 / t_{1/2})(110)} \Rightarrow t_{1/2} = 52.0 \text{ s}$ B1 for repeated reading and/or average. So a student who reads $t = 51 \text{ s}$ from the graph at $\frac{1}{2} C_0 = 32.5 \text{ s}^{-1}$ will score 3 marks (losing last B1 mark) The decaying radioactive nuclide is $^{220}_{86}\text{Rn}$. A0	
		(ii)		
	(b)		At $t = 120 \text{ s}$, $C = 14 \text{ s}^{-1}$ so count rate of Y = $65 - 14 = 51 \text{ s}^{-1}$ B1 (can be for other points but with same concept) $t = 0$, count rate of Y = 0 $t = t_{1/2}$, count rate of Y = $\frac{1}{2} C_0$ B1 for both points plotted with smooth exponential curve drawn throughd	
	(c)		$^{220}_{86}\text{Rn} \rightarrow ^4_2\alpha + ^{216}_{84}\text{Y}$ (ecf allowed)	
	(d)		linear momentum is conserved $0 = (A - 4)u v_Y + 4u v_\alpha$ M1 $(A - 4)u v_Y = 4u v_\alpha$ $(A - 4) v_Y / 4 = v_\alpha$ ratio $\frac{\text{initial kinetic energy of } \alpha\text{-particle}}{\text{initial kinetic energy of daughter nucleus Y}} = \frac{1/2 (4u) v_\alpha^2}{1/2 (A - 4)u v_Y^2} = \frac{(4) ((A - 4) v_Y / 4)^2}{(A - 4) v_Y^2}$ $= \frac{(4) (A - 4)^2}{(A - 4)(4)^2} = \frac{1}{4} (A - 4) = \frac{1}{4} A - 1$	

8

Section B

Answer **one** question in this section in the spaces provided.

7	(a)	(i)	displacement continued at where there are no beams	
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